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Tokenisation of Real-World Asset (RWA): Emerging Practices, Case Studies, and Regulatory Trends in Asia

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Abstract: This article examines the rapid growth of Real-World Asset (RWA) tokenisation in Asia, focusing on Hong Kong as an emerging regional hub. It analyses three sectoral case studies in renewable energy, real estate, and financial instruments to illustrate the practical applications, market implications, and regulatory challenges of RWA projects. As of September 2025, the global RWA market reached an estimated value of \$30.91 billion and is projected to grow into a trillion-dollar market within the next decade. The article highlights Asia's proactive regulatory initiatives aimed at developing clear tokenisation standards and promoting the sustainable and responsible growth of the virtual asset sector. Supported by regulatory sandboxes and institutional participation in leading financial centres such as Hong Kong and Singapore, the region has become a focal point of innovation in asset tokenisation. Following the introduction, Section 2 reviews the latest developments in RWA as a fast-emerging area of financial and legal practice. Section 3 presents three case studies, while Section 4 provides practical guidance for asset owners and investors. Section 5 discusses key regulatory models and the overseas expansion of Chinese enterprises through digital assets tokenisation, and Section 6 concludes with implications for regulators, investors, and policymakers.

Keywords: Real-World Asset (RWA), tokenisation, fintech regulation, Hong Kong, digital assets, financial innovation



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1. Introduction

Real-World Asset (RWA) has become a popular financial practice in today's international capital markets. As of September 2025, the total global RWA market was worth \$30.91 billion, which is on track to become a trillion-dollar market over the next decade.¹ It represents the latest development in the global fintech revolution, integrating technological breakthroughs such as artificial intelligence (AI), blockchain, data analytics, and cloud computing into the financial services sector.² RWA refers to the process of digitising and tokenising real-world assets (such as real estate, photovoltaic power plants, carbon credits, bonds, accounts receivable, and artworks) through blockchain technology, so that ownership registration, circulation, and trading can be realised on-chain.³ The practices suggest the latest financial innovation in global capital markets, which combines the digitisation of real assets with integration into existing financial markets.

The RWA tokenisation market in Asia has grown rapidly, with the world's first RWA registry platform launched in Hong Kong in August 2025.⁴ Asia's regulatory initiatives aim to develop clear tokenisation standards and promote the sustainable and responsible expansion of its rapidly growing virtual asset sector. The growth has been fueled by supportive regulatory sandboxes and institutional adoption in the region's leading financial hubs such as Hong Kong and Singapore.⁵ Major banks and asset

¹ RWA.xyz, "Analytics on Tokenized Real-World Assets". Available at: <https://app.rwa.xyz/>

² Lerong Lu, "Reconceptualising Fintech: Technological Innovations in International Finance, Commercial Applications, and Legal Issues" (2024) 39 *Journal of International Banking Law and Regulation* 14.

³ World Economic Forum, "Asset Tokenization in Financial Markets: The Next Generation of Value Exchange" (May 2025). Available at <https://www.weforum.org/publications/asset-tokenization-in-financial-markets-the-next-generation-of-value-exchange/>

⁴ Li Xiaoyun, "World's first RWA registry platform launched in Hong Kong" (China Daily, 7 August 2025), available at <https://www.chinadailyhk.com/hk/article/617560>

⁵ Hong Kong Monetary Authority, "Fintech Supervisory Sandbox (FSS)", available at <https://www.hkma.gov.hk/eng/key-functions/international-financial-centre/fintech/fintech-supervisory-sandbox-fss/>;

managers are piloting tokenised bonds, funds, and treasury products to enhance liquidity and transparency. Increasing policy clarity and the cross-border interoperability of regional fintech ecosystems are positioning Asia as a leading region for large-scale RWA applications.⁶ Against this background, the article examines the latest RWA practices in Asia, with a particular focus on Hong Kong, through an analysis of three real-world cases of RWA tokenisation. It further explores the emerging RWA models and regulatory trends in the region’s leading financial centre, offering valuable insights for financial lawyers, investment bankers, asset owners, investors, and regulators worldwide.

The article proceeds as follows. Following this introduction, Section 2 examines the latest developments in RWA as a fast-emerging area of financial and legal practice. Section 3 presents three real-world RWA case studies in Asia, focusing on the renewable energy sector, the real estate sector, and tokenised money market funds. Section 4, based on the lessons from RWA case studies, offers practical guidance for asset owners and investors seeking to participate in future RWA activities. Section 5 outlines the key RWA models and regulatory trends in Hong Kong, as well as the overseas development trend of Chinese enterprises, and Section 6 concludes the paper.

2. Global developments of RWA

and Monetary Authority of Singapore, “Overview of Regulatory Sandbox”, available at <https://www.mas.gov.sg/development/fintech/regulatory-sandbox>

⁶ Asia is a leader in large fintech experiments. For example, the mBridge project is an Asia-led multi-CBDC platform enabling real-time, peer-to-peer cross-border payments and foreign exchange using central bank digital currencies. Built on the mBridge Ledger blockchain, it ensures compliance with local regulatory and governance standards. Led by the Hong Kong Monetary Authority, Bank of Thailand, Central Bank of the UAE, People’s Bank of China, and the BIS Innovation Hub (Hong Kong Centre)—with the Saudi Central Bank joining in 2024—it positions Asia at the forefront of fintech innovation. See BIS, “Project mBridge reached minimum viable product stage” (11 November 2024), available at https://www.bis.org/about/bisih/topics/cbdc/mcbdc_bridge.htm.

RWA practices denote the tokenisation of tangible and financial assets such as real estate, bonds and commodities on blockchain networks to enhance liquidity, transparency, and regulatory compliance within the digital economy.⁷ RWAs are distinguished by three characteristics that define their transformative potential within blockchain-based financial systems.⁸ The first is ownership verification, achieved through technologies such as blockchain notarisation and zero-knowledge proofs, which ensure the authenticity, traceability, and uniqueness of on-chain assets. The second is liquidity enhancement, whereby traditionally illiquid assets such as real estate or infrastructure can be tokenised, fractionalised, and traded with reduced entry barriers. The third characteristic is cross-border accessibility.⁹ By combining regulatory compliance frameworks and programmable smart contracts, RWAs facilitate the seamless participation of global investors and enable the efficient transnational flow of capital.

Due to the advantages of RWAs, they have attracted participation from leading financial institutions and technology giants, which in turn, drives RWAs from the experimental stage towards large-scale application. Traditional financial players are increasingly recognising the potential of tokenisation to enhance liquidity, transparency, and efficiency in asset management. For instance, BlackRock has

⁷ *The Financial Times*, “How tokenisation and blockchain are shaping the future of investment”, available at <https://www.ft.com/partnercontent/defi-technologies/how-tokenisation-and-blockchain-are-shaping-the-future-of-investment.html>

⁸ Blockchain is a decentralised public ledger that records and stores transactional data across a network of nodes. The information is updated collaboratively—typically by miners—and remains visible to all participants. In principle, this distributed system of data storage and access operates without ownership or control by any single entity. See Lerong Lu, *Global Fintech Revolution: Practice, Policy, and Regulation* (Oxford University Press, 2024), p. 22.

⁹ Financial Stability Board, “Recommendations to Promote Alignment and Interoperability Across Data Frameworks Related to Cross-border Payments: Final report” (12 December 2024), <https://www.fsb.org/2024/12/recommendations-to-promote-alignment-and-interoperability-across-data-frameworks-related-to-cross-border-payments-final-report/>.

announced the launch of its first RWA fund.¹⁰ Similarly, J.P. Morgan has successfully implemented tokenised U.S. Treasury transactions on its Kinexys platform, demonstrating how established financial infrastructures can integrate blockchain-based assets.¹¹ Goldman Sachs is also actively exploring the tokenisation of money market funds, further validating the growing convergence between traditional finance and digital asset innovation.¹² Clearly, institutional players have been the driving force behind the tokenisation of traditional financial assets such as bonds, funds, and treasuries, bringing greater credibility, liquidity, and regulatory engagement to global financial markets. Meanwhile, technology firms, especially BigTechs, are advancing blockchain infrastructure and compliance solutions, fostering interoperability and accelerating the institutionalisation and mainstream use of RWAs across Asia's regional financial systems. Some major Chinese technology companies have explored RWA tokenisation as part of their fintech innovation strategies. For instance, Ant Group and JD.com launched pilot RWA projects in Hong Kong, but were subsequently instructed by mainland Chinese authorities to suspend these initiatives amid tightening regulatory oversight of digital assets.¹³

Regulatory support has also accelerated the development of RWAs, as global financial hubs work to establish clear and supportive frameworks for compliant innovation. The law, policy, and regulation

¹⁰ Natalia Karayaneva, "First Web3 Integrations With Blackrock And The RWA Trend" (13 June, 2024) Forbes. Available at: <https://www.forbes.com/sites/nataliakarayaneva/2024/06/11/first-web3-integrations-with-blackrock-and-the-rwa-trend/>

¹¹ J.P. Morgan, "Kinexys Pilots First USD-Denominated Deposit Tokens". Available at: <https://www.jpmorgan.com/payments/newsroom/kinexys-usd-digital-deposit-tokens>

¹² Goldman Sachs, "BNY and Goldman Sachs Launch Tokenized Money Market Funds Solution" (23 July, 2025). Available at: <https://www.goldmansachs.com/pressroom/press-releases/2025/bny-goldman-sachs-launch-tokenized-money-market-funds-solution>

¹³ Xinmei Shen, "China's pause on stablecoin projects will not dampen Hong Kong's crypto push, experts say" (South China Morning Post, 25 October 2025), available at <https://www.scmp.com/tech/blockchain/article/3330221/chinas-pause-stablecoin-projects-will-not-dampen-hong-kongs-crypto-push-experts-say>

of fintech play a vital role in fostering the growth of the fintech sector and promoting financial innovation within a country.¹⁴ The key regulatory dimensions of fintech include the use of RegTech to enhance supervision and compliance through real-time data, algorithms, and AI; the adoption of regulatory sandboxes that enable innovation through controlled experimentation; and the maintenance of consumer protection and financial stability as core objectives guiding the development of fintech regulation and policy. Hong Kong, Singapore, and Dubai have each introduced regulatory sandboxes that allow controlled experimentation with RWA projects under official supervision.¹⁵ These initiatives not only encourage innovation but also help mitigate risks through structured oversight. At the same time, the European Union’s Markets in Crypto-Assets (MiCA) regulation has explicitly incorporated RWAs into its virtual asset framework, creating a unified approach to governance and compliance across member states.¹⁶

The market outlook for RWAs is particularly promising. Forecasts from Boston Consulting Group (BCG) and ADDX suggest that the tokenised RWA market could reach as much as USD 16 trillion by 2030, representing approximately 10 per cent of global GDP.¹⁷ This rapid expansion is expected to be driven primarily by core sectors, including real estate, carbon credits, bonds, and supply chain finance, each offering substantial opportunities for digital transformation and capital efficiency. As these developments unfold, RWAs are poised to redefine the landscape of global finance by bridging the gap between traditional assets and blockchain-based ecosystems. By enabling on-chain

¹⁴ Lerong Lu, *Global Fintech Revolution: Practice, Policy, and Regulation* (Oxford University Press, 2024), p. 45.

¹⁵ Ibid.

¹⁶ ESMA, “Markets in Crypto-Assets Regulation (MiCA)”. Available at: <https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/markets-crypto-assets-regulation-mica>

¹⁷ BCG and ADDX, “Relevance of on-chain asset tokenization on ‘crypto winter’”. Available at: <https://web-assets.bcg.com/1e/a2/5b5f2b7e42dfad2cb3113a291222/on-chain-asset-tokenization.pdf>

representation of real-world properties, they enhance market efficiency, transparency, and accessibility for both institutional and retail investors across the globe. As regulatory frameworks mature and interoperability improves, RWAs could become a cornerstone of the next generation of digital financial infrastructure, which combines conventional finance with decentralised innovation.

The global developments of RWA have shown certain clear trends: There is a growing institutionalisation of the sector, as the focus shifts from Web3-native projects to initiatives led by banks, funds, and even central banks.¹⁸ Also, there is an increasing emphasis on regulatory compliance. Projects lacking legal and supervisory endorsement are less likely to access mainstream financial markets. The RWA landscape is undergoing progressive fragmentation, as a wider range of non-standard assets, such as intellectual property (IP), advertising traffic, and accounts receivable, are tokenised and channelled into capital markets. A growing number of practical examples of RWA applications are being observed across the US, Europe, Africa, and Asia, reflecting the global momentum of asset tokenisation.¹⁹ These developments span diverse sectors such as real estate, commodities, renewable energy, and financial instruments, demonstrating how blockchain-based tokenisation is reshaping traditional asset ownership and investment models at an unprecedented speed.

3. RWA case studies in Asia

¹⁸ Web3 represents the next evolution of the internet, powered by blockchain technology and characterised by decentralisation and user ownership. It shifts control from centralised platforms to individuals, enabling new forms of digital interaction, finance, and asset ownership. See McKinsey, “What is Web3?” (10 October 2023), available at <https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-web3>

¹⁹ Evgeny Tarasov, “Africa Launches First Tokenized Investment Platform” (Coinspaidmedia, 2 June 2025), available at <https://coinspaidmedia.com/news/africa-launches-first-investment-rwa-platform/>

As tokenisation continues to develop, a growing number of practical applications have demonstrated its transformative potential across diverse industries. With an estimated US\$11 trillion in private wealth and a US\$2.2 trillion corporate financing gap, Asia holds vast potential for RWA tokenisation to unlock liquidity, enhance capital efficiency, and promote financial inclusion across the region.²⁰ In particular, the renewable energy industry, real estate sector, and financial sector have emerged as frontrunners in adopting tokenisation models that bridge traditional assets with digital infrastructure. Examining these cases demonstrates the technological feasibility and economic value of tokenised RWAs. It also offers valuable insights into the changing regulatory and operational frameworks supporting their implementation.

The following sections present three illustrative cases of RWA tokenisation in Asia, each demonstrating distinct technological mechanisms, market impacts, and regulatory or policy implications within their respective industries. Collectively, these cases offer valuable insights into how Asia's leading financial centres are approaching the tokenisation of real-world assets through experimentation, regulatory adaptation, and public-private collaboration. They reflect Asia's unique blend of rapid technological adoption, strong regulatory oversight, and financial innovation, which together provide important lessons for balancing innovation with investor protection and market integrity. The Asian experience emphasises the need for clear regulatory frameworks, interoperability standards, and cross-border cooperation to facilitate the sustainable and inclusive growth of RWA markets in the global digital economy.

3.1 RWA in renewable energy industry: Charging-pile revenue-right tokenisation by Longshine

²⁰ Rehan Ahmed, "Building the backbone: Asia's race to realize RWA markets" (Mexc, 7 August 2025), available at <https://www.mexc.com/en-PH/news/building-the-backbone-asias-race-to-realize-rwa-markets-opinion/64111>.

The project is run by Longshine Technology Group, a Chinese company that offers digital energy services and smart-energy solutions.²¹ In this project, the asset that is tokenised is the usage-rights and revenue-rights of electric-vehicle charging stations (charging piles). The structure focuses on part of the future fee income of those charging units. The charging-pile future charging-fee revenue is mapped onto a blockchain, and the RWA product is issued. The purpose of the financing is to expand the charging-pile construction funding sources, ease the pressure of initial capital investment, and test an RWA model in the green-energy-consumption scenario at the same time.

The project was developed within the HKMA's 'Project Ensemble' sandbox,²² which provides a controlled environment for financial innovation under clear regulatory supervision. By partnering with a licensed digital-asset firm, Longshine ensured that the token structure, issuance, and distribution followed existing financial rules in Hong Kong. The pilot is currently open only to qualified institutional investors, which reflects a cautious approach to investor protection. It is not accessible to retail investors in mainland China. Despite these limits, the case carries strong policy significance. It expands the practical use of RWA beyond traditional financial or real-estate assets and demonstrates how green-transportation infrastructure can be integrated into the digital finance ecosystem. The model also sets a precedent for tokenising public-facility assets in the future, such as city-level charging networks or renewable-energy installations.

This tokenisation of charging-pile revenue-right marks a clear departure from traditional power-station revenue-rights models. It focuses on individual consumers and daily green consumption rather than large, centralised energy assets. By tokenising the use-rights and fee-revenue of charging piles, it

²¹ Longshine, "About Longshine". Available at: <https://en.longshine.com/about>

²² Longshine Technology Group Co.,Ltd, "Annual Report 2024 (Summary)". Available at: <https://en.longshine.com/upload/file/2025/0423/1745394201540338.pdf>

brings smaller and decentralised energy-consumption activities onto the blockchain for the first time. In doing so, it links the concept of green transportation with practical financial innovation. The project ties carbon reduction directly to real consumer behaviour in the electric-vehicle ecosystem, therefore turning sustainable mobility into a measurable and investable asset. At the same time, it introduces a user-interactive design that turns part of the tokenised rights into ‘charging-benefit vouchers’. These vouchers can be used both as an investment instrument and as a form of consumption credit, which allows token holders to gain financial returns while also enjoying the tangible benefits of green travel.

The Longshine project builds a clear link between real-world energy use and digital financial assets. For the company, tokenisation offers a way to turn its expected charging-fee income into immediate capital. This helps reduce the heavy upfront investment needed for building charging stations and supports faster network expansion. Instead of waiting years to recover costs through gradual charging fees, Longshine can now raise funds directly from the market by selling tokenised rights to those future revenues. Investors, on the other hand, gain exposure to a new type of income-generating asset. By holding the tokens, they can receive stable cash flow from actual charging-fee revenues, while also obtaining green-mobility consumption rights of sustainable transport. The process is simple and transparent: charging generates income, an SPV secures and records those rights, tokenisation divides the asset into digital units, and investors hold these tokens to receive returns and usage benefits. In essence, the model turns everyday charging activities into tradable on-chain certificates that combine both investment yield and real consumption value.

This case shows that public-service assets like charging stations are naturally suited for RWA development. Their cash flows are frequent, measurable, and supported by reliable data, making them easier to track and tokenise. Green mobility has become a central focus of ESG investment, and projects that combine financial innovation with environmental value can attract both institutional and

policy interest. Companies such as Longshine, which already operate digital energy platforms, are in a strong position to lead such initiatives, since they control both the data and the transaction environment needed for tokenisation.

However, challenges still remain. The charging-fee rights are small and spread across many users, so gathering and splitting them efficiently on-chain poses technical difficulties. The income level is also closely tied to how often electric vehicles are used, which makes the returns somewhat unstable. In addition, the investor base for this type of hybrid product is still developing. Many potential investors are not yet familiar with combining financial returns and consumption rights in one token. Building confidence and understanding among users and investors will therefore be essential for scaling such projects in the future.

3.2 RWA in real estate industry: Tokenisation of commercial office building in Causeway Bay

In 2023, a landmark office building project in Causeway Bay Hong Kong, valued at HKD 2.9 billion, underwent tokenisation through the issuance of one million ERC-3525 tokens,²³ with each token representing an equal share of ownership in the property.²⁴ Investors could trade these tokens using a Hong Kong dollar denominated stablecoin or central bank digital currency (CBDC). Rental income from the property was distributed daily via smart contracts to ensure transparency and automation in

²³ ERC-3525 is an Ethereum standard for semi-fungible tokens (SFTs) that combines the quantitative, divisible nature of fungible tokens with the unique ID and expressivity of non-fungible tokens. See “What is ERC-3525- FROM NFTs to SFTs Thanks to a Combination of ERC-20and ERC-721” (2023) *BingX Blog*. Available at:

<https://blog.bingx.com/events/what-is-erc-3525-from-nfts-to-sfts-thanks-to-a-combination-of-erc-20-and-erc-721/#:~:text=ERC%2D3525%20is%20a%20semi,Metaverse%2C%20GameFi%20and%20DeFi%20sectors.>

²⁴ Yashika Thakur, “RWA Tokenization In Hong Kong: Paving The Way For The Future Of Asset Ownership”. Antier. Available at: <https://www.antiersolutions.com/blogs/rwa-tokenization-in-hong-kong-paving-the-way-for-the-future-of-asset-ownership/#:~:text=In%20early%202023%2C%20Tokenize%20Xchange,to%20add%20to%20their%20portfolios.>

profit allocation. Meanwhile, Internet of Things (IoT) sensors were deployed throughout the building to monitor real-time occupancy and vacancy rates, which provides continuous data verification to support asset valuation and performance tracking.

Trading took place on HKbitEX, a licensed exchange in Hong Kong, which provided 24 hour's liquidity for token holders.²⁵ The project demonstrated how blockchain technology, when combined with regulatory compliance and real-world asset integration, could revolutionise the property investment landscape by improving liquidity, enhancing transparency, and lowering capital barriers for both institutional and retail investors.

Such business model benefits the key participants in the ecosystem. For the real-estate developer, tokenisation allows early monetisation of property assets and accelerates capital circulation. Instead of waiting for slow rental returns or large-scale asset sales, the company can raise funds directly from the market through token issuance. For investors, the model lowers the barrier to entry with minimum participation starting at just HKD 1,000, while offering exposure to a stable source of rental income and the potential for capital appreciation. The structure also diversifies investment risks by allowing fractional ownership across high-value real-estate assets. For the trading platform, the project provides a compliant security-token-offering (STO) channel that expands its regulated digital-finance business and strengthens Hong Kong's position as a hub for asset tokenisation.

The tokenisation of the office building in Causeway Bay stands as one of Asia's earliest and most representative examples of real-estate tokenisation. The project may reshape the regional REITs market and serve as a model for digital transformation in property finance. For mainland Chinese property owners, especially those managing high-value but low-liquidity assets such as office buildings

²⁵ HKbitEX. Available at: <https://www.hkbitex.com.hk/en-US>

and industrial parks, Hong Kong's regulated security token offerings (STO) framework provides a clear and replicable model. The pathway of asset division, settlement in Hong Kong dollars or stablecoins, and circulation through licensed trading platforms offers a practical example of how traditional real-estate assets can access cross-border financing under a compliant and transparent structure.

3.3 RWA in financial industry: ChinaAMC RMB tokenised money fund

In July 2025, ChinaAMC (Hong Kong) announced the launch of the world's first RMB tokenised money fund.²⁶ Unlike traditional mutual funds, whose ownership is recorded through conventional securities accounts, this tokenised fund represents ownership through digital tokens stored on a blockchain. While the underlying assets remain conventional securities, such as short-term government bonds and policy bank bonds, the tokenised structure fundamentally redefines how ownership, custody, and transparency are managed. Its primary goal is to overcome foreign exchange restrictions and enable the global circulation and investment of RMB-denominated sovereign debt through a compliant offshore structure. Six distributors are participating in the launch, including major securities firms like Futu Securities, Guotai Junan Securities, CITIC Securities, and Huaying Securities, along with one asset management company, Xinglu Financial Ltd., and one licensed virtual asset trading platform, OSL Digital Securities Ltd. According to ChinaAMC (Hong Kong), licensed virtual asset exchanges frequently serve as both custodians and distributors in tokenised fund issuance, ensuring regulatory compliance while providing technological infrastructure for custody and settlement.

²⁶ Selen Li and Sneha Kumar, "China's second-largest fund house launches world's first renminbi tokenised money fund" (17 July, 2025). *Reuters*. Available at: <https://www.reuters.com/markets/asia/chinas-second-largest-fund-house-launches-worlds-first-renminbi-tokenised-money-2025-07-17/>

ChinaAMC's project draws inspiration from the BlackRock BUIDL model. In 2024, BlackRock has launched its first-ever tokenised money market fund on the Ethereum blockchain: the BlackRock USD Institutional Digital Liquidity Fund, known as "BUIDL".²⁷ The initial network of partners supporting BUIDL includes Anchorage Digital Bank NA, BitGo, Coinbase, and Fireblocks, along with other key players in the digital asset infrastructure ecosystem.²⁸ This fund enables eligible investors to earn returns tied to the U.S. dollar by purchasing shares through Securitize Markets, LLC. Designed to maintain a steady \$1 value per token, BUIDL distributes accumulated daily yields each month in the form of newly issued tokens directly to investors' digital wallets. The fund's assets consist entirely of cash, U.S. Treasury bills, and repurchase agreements, giving investors blockchain-based exposure to traditional short-term instruments while generating yield. Token holders can move their holdings at any time, with 24 hours a day year-round to other verified investors. Participants may choose from several custody options for managing their tokens.²⁹

Inspired by BUILD, ChinaAMC's RMB Tokenised money fund combines China's sovereign bonds with a RWA fund architecture for the first time. By balancing the public, immutable, and traceable nature of blockchain, every transaction and change in ownership is recorded in real time, which offers investors unprecedented transparency in portfolio holdings and fund operations. This stands in contrast to traditional funds, where reporting is often delayed or opaque. Tokenisation also broadens accessibility. Besides being distributed through conventional financial intermediaries such as banks and brokers, tokenised funds can be traded on licensed virtual asset exchanges, which enables more

²⁷ Vicky Ge Huang, 'BlackRock Launches First 'Tokenized Fund on Ethereum Blockchain' (20 March, 2024) *The Wall Street Journal*. Available at: <https://www.wsj.com/livecoverage/fed-meeting-fomc-interest-rate-decision-march-2024/card/blackrock-launches-first-tokenized-fund-on-ethereum-blockchain-nzDSjjH5mEijUzKO24T4>

²⁸ Securitize, 'BlackRock Launches Its First Tokenized Fund, BUIDL, on the Ethereum Network' (21 March, 2024). Available at: <https://securitize.io/learn/press/blackrock-launches-first-tokenized-fund-buidl-on-the-ethereum-network>

²⁹ Ibid.

flexible and efficient market participation. It also introduces 24/7 trading and instant settlement (T+0), which marks a major efficiency improvement over the traditional Qualified Domestic Institutional Investor (QDII) channels, which typically operate on a T+3 basis. The fund also bridges the functions of a stablecoin and a sovereign bond fund, since each tokenised unit is pegged to one US dollar while earning yields from Chinese government bonds. Therefore, it merges the features of a money market fund with those of fixed-income securities.

The business model reflects a clear alignment of incentives. For ChinaAMC and the sovereign issuers, the fund opens a new channel for attracting international investors and expanding RMB-denominated assets into the global capital market. For investors, the returns are derived from interest on Chinese government and policy bank bonds, offering an annualised yield of about 3.1 per cent, which is approximately 0.8 per cent higher than comparable QDII products. The structure ensures stable value (1:1 USD peg) combined with fixed income from high-grade bonds, following a transparent value chain. Investors allocate funds, the RWA fund holds tokenised sovereign debt, and income flows back directly to token holders through automated smart contracts.

From a regulatory perspective, the project operates fully under the Hong Kong legal and financial framework, while effectively bypassing the limitations of China's domestic foreign exchange controls. The fund's assets are custodied on a licensed exchange, with the underlying bonds held in physical form to ensure full 1:1 redemption security. This arrangement creates a robust compliance pathway that balances innovation with financial integrity.

The project carries important implications for both financial innovation and RMB internationalisation. It demonstrates that even high-grade sovereign assets can achieve cross-border liquidity and round-the-clock trading through RWA tokenisation, offering a new avenue for expanding the use of the RMB in global markets. However, coordination across multiple regulatory regimes is complex, secondary

market depth and liquidity are still changing. But, standing as China's first sovereign bond-based RWA funds, the initiative marks a breakthrough in applying blockchain to sovereign debt markets. By enabling 24-hour global circulation of RMB bonds and reducing settlement times from T+3 to T+0, the project has successfully attracted investment from Middle Eastern sovereign wealth funds and Singaporean family offices. It also represents a milestone for both RMB asset internationalisation and the practical deployment of RWA frameworks in mainstream financial instruments.

In addition to the RMB-denominated fund, ChinaAMC (Hong Kong) also launched a USD-denominated tokenised fund, the ChinaAMC USD Digital Currency Fund.³⁰ Together with the previously approved ChinaAMC HKD Digital Currency Fund, the company now offers a comprehensive tri-currency tokenised fund series which covers Hong Kong dollar, Chinese yuan, and US dollar. This makes ChinaAMC (Hong Kong) one of the few fund managers in Asia with a full suite of tokenised money market funds denominated in multiple major currencies.

According to the company, these three funds aim to combine efficiency, transparency, and regulatory robustness through blockchain technology.³¹ By digitising fund units, ChinaAMC (Hong Kong) has achieved significant improvements in settlement speed, liquidity management, and real-time portfolio visibility, thereby enhancing the overall investor experience. The project demonstrates Hong Kong's growing role as a regional hub for regulated digital finance, bridging the gap between traditional asset management and the emerging world of blockchain-based investment products.

Another prominent example is Robinhood, a company widely recognised in the U.S. equity market. Beyond its traditional stock trading services, Robinhood has recently entered the field of tokenised

³⁰ ChinaAMC (Hong Kong), "ChinaAMC (HK) Launches New Tokenized Funds ChinaAMC USD & RMB Digital Money Market Funds"(17 July, 2025). Available at: https://www.chinaamc.com.hk/wp-content/uploads/2025/07/ChinaAMC-HK-Launches-USD-and-RMB-Tokenized-Money-Market-Funds_EN_Final.pdf

³¹ Ibid.

equities by offering blockchain-based representations of shares.³² Although the company itself is listed on the U.S. stock exchanges, it simultaneously mints on-chain tokens corresponding to equity holdings. This approach shares certain conceptual similarities with stablecoins but serves a distinct financial function. The issuance of tokenised stocks must be backed by real equity assets. These underlying shares may derive from Robinhood's proprietary holdings or be acquired via SPVs established for this purpose. Even the shares of unlisted companies such as OpenAI and SpaceX are reportedly included in these tokenisation processes.³³ This practice produces an intriguing phenomenon: companies that remain privately held in traditional markets effectively become 'listed' on the blockchain, as their tokenised shares can circulate freely on secondary markets that do not require formal listings on NASDAQ or the New York Stock Exchange.

This development is particularly significant when contrasted with the traditional process of investing in U.S. equities, which is notoriously cumbersome for international investors. Conventionally, one must obtain a foreign bank account, convert local currency into U.S. dollars, navigate complex compliance and taxation procedures, and open a foreign brokerage account. In contrast, platforms such as Robinhood and AXIS Stocks, by tokenising equities on-chain, have greatly simplified this process. Investors now need only hold stablecoins such as USDT or USDC to purchase shares in companies like Tesla, Apple, or even unlisted firms such as OpenAI and SpaceX, all with 24-hour, T+0 settlement. This represents a remarkable leap in accessibility and efficiency.

³² Sandy Carter, "Robinhood's RWA Letter Could Bring Wall Street Onchain" (20 May, 2025) Forbes. Available at: <https://www.forbes.com/sites/digital-assets/2025/05/20/robinhoods-rwa-letter-could-bring-wall-street-onchain/>

³³ "Robinhood gives out tokens of OpenAI and SpaceX in Europe. Stock hits record" (30 June, 2025) CNBC. Available at: <https://www.cnbc.com/2025/06/30/robinhood-stock-openai-spacex-tokens.html>

In comparison with traditional stock exchanges, on-chain equities and tokenised securities arguably represent a more advanced form of financial infrastructure. In the future, high-quality corporate equity may no longer require a formal public listing to become accessible to investors. Consider, for instance, a high-performing private company whose shares are unavailable through conventional markets. Normally, only institutional investors, venture capitals, or individuals with insider connections with the founding shareholders of the company could access such opportunities. However, through tokenisation, if a company's shares are issued on-chain, ordinary investors may participate directly. This process significantly enhances equity market inclusiveness and democratises access to early-stage or high-value investment opportunities that were once reserved for elite investors. In this sense, tokenisation enables the acquisition of high-quality, illiquid equity assets through virtual asset exchanges that would otherwise be inaccessible via traditional secondary markets. This mechanism exemplifies the broader concept of RWA tokenisation by including both physical and financial assets. Through RWAs, investors can use stablecoins to gain exposure to almost any asset that could traditionally be purchased with fiat currency.

4. Practical advice for asset owners and investors

A central issue in the development of RWA tokenisation lies in defining the legal connection between the tokenised instrument and its underlying asset. This linkage determines both the legal nature of the token, whether it constitutes an ownership interest, a receivable claim, or a contractual entitlement, and the enforceability of investors' rights in the event of default or dispute. Establishing this relationship is therefore central to ensuring legal certainty, investor protection, and regulatory compliance in tokenised asset financing.

From a practical perspective, RWA tokenisation can be understood as the process of transforming productive capacity or cash-flow, which means generating rights into digitally represented financial

instruments.³⁴ For instance, a manufacturing company may place part of its annual production capacity on-chain, and issue tokens that correspond to a specific quantity of future output. Investors who purchase these tokens effectively acquire a right to a portion of the future economic benefit generated by that production line. The blockchain serves as the registry and transactional infrastructure through which these rights are recorded, transferred, and settled.

The legal and economic structure of tokenised assets can vary depending on the underlying transaction.³⁵ For instance, in one model, the tokens function as yield-bearing instruments.³⁶ Investors receive periodic income distributions tied to the production schedule or sales revenue of the underlying asset. This approach contrasts with traditional equity dividends, which are typically distributed on a quarterly or annual basis through centralised intermediaries such as banks or custodians. Smart contracts embedded in the blockchain enable automated and rule-based distribution. The continuous visibility of production and transaction data also reduces the risk of information asymmetry and minimises opportunities for manipulation. In another model as an example, the tokens represent direct claims to tangible output or performance rights.³⁷ An enterprise may, for example, tokenise the right to purchase a defined amount of renewable energy or industrial output. The buyer of the token acquires a claim to that output, and upon settlement, the corresponding physical asset is delivered and the token is retired. This model creates a direct and legally verifiable connection between

³⁴ Richard A. Brealey, Stewart C. Myers, and Franklin Allen, *Principles of Corporate Finance* (10th edition, McGraw-Hill, 2011), p. 90.

³⁵ Xavier Lavyssière, “Legal Structures of Tokenised Assets” *European Journal of Risk Regulation* (2025). doi:10.1017/err.2024.88

³⁶ Crypto.com, “Tokenisation of RWAs & Yield-Bearing Stablecoins” (7 May, 2024). Available at: <https://crypto.com/en/research/rwa-tokenisation-april-2024>

³⁷ Chainalysis, “An Introduction to ERC-3643: The Real-World Asset (RWA) Token Standard” (23 April, 2025). Available at: <https://www.chainalysis.com/blog/introduction-to-erc-3643-ethereum-rwa-token-standard/>

the token and the underlying productive asset, and therefore effectively bridging the digital and physical layers of ownership and value transfer.

Both approaches show the importance of clear legal articulation between tokenised representations and their real-world references. In most cases, this relationship is established through contractual agreements that specify the rights and obligations attached to the token, including the mechanisms of valuation, redemption, and dispute resolution. The credibility of the tokenisation process depends on whether these contractual claims can be enforced in accordance with existing legal frameworks. Jurisdictions such as Hong Kong³⁸, which are developing regulatory guidance for tokenised funds and RWA instruments, provide a foundation for legitimising this process by recognising tokenised claims as lawful representations of asset ownership or entitlement within a regulated environment.

However, not all assets are equally suitable for tokenisation. Assets best suited for on-chain representation typically possess intrinsic and measurable economic value, along with predictable cash flows that can support periodic distribution. Production capacity in high-end manufacturing and renewable energy projects are illustrative examples. These assets generate steady, quantifiable returns and are backed by operational processes that can be continuously monitored. Their underlying data are often already digitised and verifiable through automation technologies, such as IoT-enabled sensors or industrial data systems, which enhances transparency and auditability.³⁹ Furthermore, assets

³⁸ Securities and Futures Commission (Hong Kong), “A-S-P-I-Re” for a brighter future: SFC’s regulatory roadmap for Hong Kong’s virtual asset market (19 February 2025), <https://www.sfc.hk/-/media/EN/files/ER/ASPIRe/ASPIRe-roadmap-for-Hong-Kongs-virtual-asset-market-Eng.pdf?rev=1ff2b9ab976f482e924b1d911c55b27a&hash=ABF3EDC5C737FE8435568222D04417EA>

³⁹ The Internet of Things (IoT) is a network of interconnected physical devices—such as sensors, machines, and everyday objects—that collect, exchange, and act on data through the internet. By linking the physical and digital worlds, the IoT allows real-time monitoring, automation, and optimisation across sectors such as manufacturing, healthcare, transport, and smart cities. See McKinsey, “What is the Internet of Things (IoT)?” (28 May 2024), available at <https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-the-internet-of-things>

that exhibit a high degree of digitalisation and operational traceability are more likely to retain investor confidence in tokenised form. The ability to provide real-time, tamper-resistant data on output, use, or performance strengthens the reliability of the digital representation and reduces the legal and informational gap between the on-chain and off-chain environments. This combination of intrinsic value, stable cash flow, and verifiable data integrity makes certain industrial and energy assets particularly well aligned with the logic of the tokenisation of RWA.

Before starting a tokenisation of RWA project, it is important to first assess whether the asset can actually be sold or traded. This process begins with checking a series of issues. The first is to check the stability of cash flow. The asset should have a steady and predictable cash flow over at least the past twelve months or within the contract period. It is also necessary to check the ownership of the asset. The rights of the asset must be clear, free from disputes, and transferable. Data source is another aspect to care. Related data should be available in real time on-chain or be verifiable through reliable audits. These steps help confirm that the asset is ready for tokenisation and that investors can trust its quality and transparency. After the necessary security checks, the next step is to choose the appropriate legal jurisdiction. If the asset is located in mainland China and the investment funds need to return to China, Hong Kong is usually the best option because it provides a bridge between mainland and international capital.

When launching an RWA project, it is usually better to start with institutional issuance. The first round of investors should be institutions or qualified investors who can handle higher risks and meet compliance standards. Once the regulatory framework and secondary trading markets become more mature, retail investors can then be considered. This gradual approach helps build credibility and ensures long-term stability.

Technology and credit enhancement should be developed together at the same time. Combining IoT or SCADA systems with on-chain audits can make asset performance more transparent.⁴⁰ Adding optional insurance or bank guarantees can further improve investor confidence. This combination makes the project more acceptable to institutional investors who value both transparency and risk control. It is also important to design a clear exit or liquidity plan. The project should define how secondary trading will work, whether through a licensed exchange, a market-making pool, or a custodian-based redemption mechanism. This would ensure that investors can trade or exit their positions when needed, which is essential for market confidence. Finally, compliance and tax planning must come before any actual launch. Before finalising the project, legal, tax, and foreign exchange experts should conduct a joint review across all involved jurisdictions, especially if the project spans multiple countries. This helps prevent future problems such as blocked fund transfers or tax disputes.

The next wave of RWA growth will not happen across all asset types at once. It will start with institutional, standardised, and data-verifiable assets. Sectors like renewable energy, real estate, and financial capital are most likely to achieve large-scale development. Other industries will need to wait until technology, regulation, and market understanding become more mature.

5. RWA models and regulatory trends

5.1 Models and regulatory trends in Hong Kong

As seen in the above three cases, the tokenisation of RWAs does not follow a single blueprint. Whether it is tied to renewable energy projects, property markets, or financial capital, its form is closely linked to the underlying asset. Each of these categories carries its own features: some focus on long-term infrastructure investment, others revolve around tangible ownership rights, while still others depend

⁴⁰ Aliyu Enemosah and Ogbonna George Ifeanyi, “Cloud security frameworks for protecting IoT devices and SCADA systems in automated environments” (2024) 22 *World Journal of Advanced Research and Reviews* 2232.

on compliance with sustainability standards. RWA tokenisation cannot succeed as a one-size-fits-all solution, the key lies in adapting it to the unique characteristics of each asset class.⁴¹

In the renewable energy sector, tokenisation has opened new pathways for financing their capital-intensive infrastructure, ranging from solar farms and offshore wind parks to emerging green hydrogen facilities. By issuing digital tokens that act like small shares, projects can now be funded by a wider pool of participants, not just large institutions. The RWA model offers a high level of standardisation with relatively stable returns, which is supported by long-term contracts or revenue from electricity or usage fees. The challenge is ensuring tokens accurately represent real-world energy generation and revenue, such as electricity delivered to the grid, in ways that are both legally binding and technologically secure.

The same tension between on-chain innovation and off-chain enforcement arises in more established asset classes such as real estate, where tokenisation promises to tackle the chronic problem of illiquidity. By converting ownership rights in residential or commercial property into a fixed number of digital tokens, investors can acquire or divest fractional shares without resorting to traditional structures that are often complex and costly. Such transformation mainly depends on the enforceability of claims, usually achieved through SPVs that hold title to the property on behalf of token holders. Regulators face the challenge of deciding whether such tokens should be treated as securities, as instruments of title, or as a hybrid form implicating both property and financial law.

If property tokenisation digitises physical assets, financial tokenisation goes further by converting capital itself into digital form. Tokenising debt instruments, such as corporate bonds, treasury bills,

⁴¹ RWA tokenisation requires an asset-specific approach, as different asset classes present distinct legal, valuation, and liquidity challenges. While standardised financial instruments may be easier to tokenise within existing legal and regulatory frameworks, illiquid or unique assets demand tailored regulatory and technological solutions to ensure accurate valuation, compliance, and market integrity.

and private credit products, is mainly about efficiency, not access. Moving these instruments onto blockchain allows for instant transactions, lower default risk, and automatic processes like interest payments and redemptions.

Among offshore jurisdictions, Hong Kong offers the most policy-friendly environment for Mainland assets, by balancing international compliance with regulatory compatibility at home. The regulatory framework is shaped by multiple authorities, including the Securities and Futures Commission (SFC), the Hong Kong Monetary Authority (HKMA), and the Hong Kong government. Its strategic direction is articulated through documents such as the Policy Statement 2.0 on the Development of Digital Assets in Hong Kong.⁴² Alongside these institutional pillars, Hong Kong has also promulgated the Stablecoins Ordinance which came into force on 1 August 2025.⁴³ The law institutes a complete regulatory system for issuing, offering, and marketing fiat-referenced stablecoins, with the aim to create a regulatory foundation for broader RWA applications. Apart from complementing the formal rules, the HKMA also launched the Project Ensemble Sandbox to facilitate testing of RWA applications and their settlement processes.⁴⁴ This Sandbox is a key implementation pillar of the wider Project Ensemble, which was initiated in March 2024 with the goal of building a new financial market infrastructure (FMI) to enable interbank settlement of tokenised assets via wholesale central bank

⁴² Financial Services and the Treasury Bureau. Full text available at:

https://gia.info.gov.hk/general/202506/26/P2025062600269_500089_1_1750909574183.pdf

⁴³ Hong Kong Monetary Authority, “Regulatory Regime for Stablecoin Issuers”. Available at:

<https://www.hkma.gov.hk/eng/key-functions/international-financial-centre/stablecoin-issuers/>

⁴⁴ Hong Kong Monetary Authority, “HKMA launches Project Ensemble Sandbox to accelerate adoption of tokenisation” (28 August, 2024). Available at: <https://www.hkma.gov.hk/eng/news-and-media/press-releases/2024/08/20240828-3/>

digital currency (wCBDC).⁴⁵ The guiding idea is not merely compliance in isolation but compliance in a way that ensures continued connectivity with Mainland China.

In practice, this policy framework has translated into concrete transaction pathways. Mainland assets are frequently channelled through SPVs incorporated in Hong Kong, which then operate under established foreign exchange and outbound investment regimes. One such pathway is the Outbound Direct Investment (ODI) scheme, which provides a regulatory basis for domestic enterprises to invest abroad.⁴⁶ Another is the Notice on Relevant Issues concerning Foreign Exchange Administration for Domestic Residents Engaging in Overseas Financing and Investing through Round-Trip Investment via Special Purpose Companies (“Circular 37”) issued by China’s State Administration of Foreign Exchange (SAFE).⁴⁷ By combining these channels, capital can be transferred outward from China, structured and raised in Hong Kong within a compliant framework, and subsequently reintegrated into the Mainland economy through legally recognised mechanisms. The loop satisfies both the requirements of Mainland foreign exchange control and the expectations of investors. The pathway, however, is not without friction. It requires formalities such as foreign exchange registration and round-trip investment filings, which introduce both procedural complexity and regulatory oversight. Nevertheless, the model has already been tested in practice. Notable RWA projects examples in Hong Kong include the cooperation of Ant Digital and Longshine Technology, which demonstrate how the

⁴⁵ Hong Kong Monetary Authority, “HKMA unveils Projects Ensemble to support the development of the Hong Kong tokenisation market” (7 March, 2024). Available at: <https://www.hkma.gov.hk/eng/news-and-media/press-releases/2024/03/20240307-5/>

⁴⁶ Clifford Chance Briefing note, “China relaxes outbound direct investment rules” (June 2014). Available at: <https://www.cliffordchance.com/content/dam/cliffordchance/briefings/2014/06/china-relaxes-outbound-direct-investment-rules.pdf>

⁴⁷ Full text available at: <https://www.safe.gov.cn/en/2014/1128/749.html>

framework can be mobilised to channel Mainland-origin assets into a globalised financial environment.⁴⁸

What advantages does this approach offer? The most prominent benefit is how well it accommodates Mainland assets. Unlike many offshore jurisdictions, Hong Kong allows structures that can eventually be reintegrated into the Mainland's legal and financial order. This ability to connect domestic regulation, while still offering access to international capital, makes Hong Kong attractive for products designed for both Mainland and overseas institutional investors. However, these advantages come with costs. The licensing and compliance thresholds in Hong Kong remain demanding. Operating an RWA platform is treated as a regulated activity, and the specific obligations depend heavily on the platform's business model. In many cases, multiple licenses from the SFC or the HKMA are required, covering activities such as dealing in virtual assets, providing custody services, or issuing stablecoins. Each license carries distinct compliance responsibilities.⁴⁹ A strict anti-money laundering and counter-financing of terrorism (AML/CFT) framework is another demanding formality for the platforms. Comprehensive Know-Your-Customer (KYC) and Customer Due Diligence (CDD) procedures must be applied to all users, while sophisticated on-chain transaction monitoring systems are necessary to identify and report suspicious activities promptly.⁵⁰ Deploying RegTech solutions to automate monitoring processes has become almost indispensable, but these tools further increase operational

⁴⁸ Wang Hanyu, "In Hong Kong, have enterprises going global found a new 'source of wealth'?" (7 November, 2024) 36 Kr. Available at: <https://eu.36kr.com/en/p/3025644074329603>

⁴⁹ Financial Services and the Treasury Bureau. "Government's Policy Statement 2.0 on Development of Digital Assets in Hong Kong" (June 26, 2025) Available at: <https://sc.isd.gov.hk/TuniS//www.info.gov.hk/gia/general/202506/26/P2025062500847.htm?fontSize=1>

⁵⁰ Hong Kong Monetary Authority, "e-HKD: A Policy and Design Perspective". Available at: https://www.hkma.gov.hk/media/eng/doc/key-functions/financial-infrastructure/e-HKD_A_Policy_and_Design_Perspective.pdf

complexity and cost.⁵¹ The demands of compliance extend into the field of data governance under Hong Kong's Personal Data (Privacy) Ordinance (PDPO). Platforms have to explore the balance between the traceability required by AML/CFT rules and the protection of individual privacy.⁵² Legal frameworks must clearly define what data may be collected and for what purposes, while technical solutions such as pseudonymous addresses or zero-knowledge proofs allow verification of transactions without exposing sensitive user identities.⁵³ These requirements illustrate the dual nature of Hong Kong's regulatory environment: it confers legitimacy and builds investor confidence, but it also imposes operational and technological burdens that only well-capitalised platforms with strong institutional backing can realistically manage.

5.2 Mainland China enterprises 'Going Global': From product export to asset tokenisation

The international expansion of Mainland Chinese enterprises has traditionally been driven by the export of goods, services, and production capacity. However, these enterprises often encounter substantial liquidity constraints in the process of globalisation. Structural mismatches between slow revenue inflows and rapid expenditure outflows remain a persistent challenge. Extended payment collection periods, long production and logistics cycles, and protracted capital turnover in cross-border transactions all contribute to financing pressure of the companies. The geographic dispersion of overseas supply chains and the opacity of counterpart creditworthiness make it even more difficult to obtain traditional bank credit, which further constrains liquidity problem of the company.

⁵¹ Medium, "Building on Bedrock: Audit and Compliance in Hong Kong's RWA Tokenization Era" (11 July, 2025). Available at: <https://achworldwide.medium.com/building-on-bedrock-audit-and-compliance-in-hong-kongs-rwa-tokenization-era-cbd66ca5a124>

⁵² Ibid.

⁵³ Ibid.

In recent years, the notion of ‘going global’ has began popular among companies in Mainland China.⁵⁴ The next phase of internationalisation for Chinese enterprises will extend beyond the outward movement of goods, services, and human capital, but toward the internationalisation of capital and assets. This transformation can be understood as a progressive sequence from product export to technological export, then capital export, and ultimately asset export. Within this trajectory, RWA tokenisation has been as a key mechanism which enables the global circulation of Chinese enterprise assets.

As previously discussed in detail, RWA tokenisation allows firms to convert real-world financial and physical assets, such as receivables, inventory, and short-term debt instruments, into on-chain digital representations. Through blockchain technology, tokenisation enhances transparency, and traceability, therefore offering new liquidity channels which transcend traditional financial boundaries. For Mainland companies facing structural financing constraints, RWA mechanisms provide an alternative approach to unlocking asset value and participating in global capital markets.

This development is particularly relevant in the context of Hong Kong’s regulatory framework for tokenised funds and digital assets. As Hong Kong positions itself as a compliant and internationally recognised hub for digital finance, it serves as a critical bridge linking Mainland enterprises with global investors and institutional liquidity.⁵⁵ The regulatory recognition of RWA products in Hong Kong thus not only supports financial innovation but also facilitates the asset internationalisation of Chinese enterprises in a lawful, transparent, and globally interoperable manner. In this sense, RWA

⁵⁴ Willa Wu and Lam Ka-sing, “Hong Kong launches task force to help mainland Chinese firms go global” (6 October, 2025) *South China Morning Post*. Available at: <https://www.scmp.com/news/hong-kong/hong-kong-economy/article/3327968/hong-kong-launches-task-force-help-mainland-chinese-firms-go-global>

⁵⁵ Dong Yu, “Hong Kong a launchpad to empower mainland companies’ overseas expansion” (27 October, 2025) *China Daily*. Available at: <https://www.chinadailyhk.com/hk/article/622472>

tokenisation represents more than a technological advancement, it constitutes an extension of China's broader 'going global' strategy. By enabling the visibility and mobility of digitalised assets on a global scale, it marks the transition from a trade-driven model of internationalisation to one rooted in financial and asset integration.

For mainland Chinese enterprises exploring the tokenisation of RWA, compliance considerations extend far beyond the technical implementation of blockchain infrastructure. They include the entire legal ecosystem governing cross-border capital flows, data security, securities regulation, and legal enforceability. Each of these dimensions presents potential regulatory risks that must be addressed through deliberate structuring and proactive legal design to ensure the legitimacy of RWA tokenization activities both within and outside China.

A primary area of concern is the legality of cross-border capital movements. When enterprises issue RWA tokens to overseas investors or repatriate proceeds from token sales, the movement of funds across borders must comply with foreign exchange control requirements and financial supervision standards. The capital raised through token issuance and the subsequent flow of investment returns are subject to the scrutiny of both domestic and foreign regulators. Accordingly, enterprises must ensure that all transactions occur through authorised and compliant payment channels, in line with the SAFE and PBoC regulations as mentioned above. Any circumvention of these rules risks violating China's capital control framework and could expose both issuers and investors to enforcement actions.

Another crucial issue relates to the protection of data security and information compliance. Tokenisation often requires the transmission of asset-related data, such as production metrics, financial performance, or operational outputs to public blockchains or foreign jurisdictions for verification and disclosure purposes. However, uploading such information across borders may trigger obligations under China's Data Security Law (DSL) and the Personal Information Protection Law

(PIPL).⁵⁶ These statutes impose strict requirements on data localisation, risk assessment, and cross-border transfer authorisation. Therefore, before uploading any data concerning domestic assets to a public or foreign blockchain, enterprises must evaluate whether such disclosures might involve sensitive industrial data, state secrets, or personal information. Compliance with data protection laws is essential not only to avoid administrative penalties but also to maintain the legal validity of the tokenisation structure itself.

Licensing and securities regulation must also be addressed. Many RWA tokens, depending on their economic characteristics, may be classified as securities or investment instruments under prevailing regulatory standards. If the token conveys profit-sharing rights, ownership claims, or debt-like entitlements, it may fall within the legal definition of a security in the jurisdiction of issuance. Consequently, companies must conduct issuance and promotional activities through licensed trading platforms or regulated exchanges, and ensure compliance with applicable securities laws. Failure to do so risks contravening prohibitions against unauthorised public fundraising, illegal securities issuance, or collective investment schemes. In practice, this means that Mainland issuers often collaborate with licensed offshore platforms, such as those based in Hong Kong to ensure that their RWA products are distributed under recognised regulatory frameworks.⁵⁷

6. Conclusion

This article has examined the landscape of RWA tokenisation in Asia, with Hong Kong emerging as a regional leader in regulatory innovation and market development. Through three case studies in the renewable energy, real estate, and financial sectors, it has shown how tokenisation can improve

⁵⁶ Rogier Creemers, “China’s emerging data protection framework” (2022) 8 *Journal of Cybersecurity* 1

⁵⁷ Li Xiaoyun, “Experts: HK tokenization shifts from pilots to wider adoption” (23 September, 2025) *China Daily HK*. Available at: <https://www.chinadailyhk.com/hk/article/620474>

liquidity, efficiency, and transparency in asset markets while introducing new legal and regulatory challenges. Moreover, as Chinese enterprises expand globally, they often encounter liquidity and financing challenges despite having solid business fundamentals, confirmed orders, and long-term contracts.⁵⁸ Traditional financing channels, such as securities issuance, equity investment, and bank lending are frequently constrained by high compliance costs, lengthy approval processes and limited flexibility. Even when viable business opportunities exist, capital mobilisation remains difficult due to structural inefficiencies in cross-border credit evaluation and the fragmented nature of international financial systems.

Typical scenarios illustrate these obstacles vividly. Enterprises establishing overseas production facilities face heightened working capital needs, as incomplete local supply chain networks and logistics inefficiencies force them to increase inventory levels and advance procurement. Mainland China enterprises serving major overseas clients are often subject to extended payment cycles and higher foreign exchange risks, which create a mismatch between cash inflows and operational outflows.⁵⁹ Meanwhile, the broader international environment characterised by trade frictions, tariff barriers, and fluctuating raw material costs continues to worsen financial uncertainty.⁶⁰ All these factors contribute to persistent liquidity pressures and rising operational costs for Chinese enterprises pursuing overseas business.

⁵⁸ UBS, “Chinese firms accelerating overseas expansion” (22 October, 2024). Available at:

<https://www.ubs.com/global/en/investment-bank/insights-and-data/2024/overseas-expansion.html>

⁵⁹ He Huifeng, “Small Chinese exporters face hit to profits as buyers demand extended payment terms” (26 October, 2025) *South China Morning Post*. Available at: <https://www.scmp.com/economy/china-economy/article/3330224/small-chinese-exporters-face-hit-profits-buyers-demand-extended-payment-terms>

⁶⁰ “Why China is winning the trade war” (23 October, 2025) *The Economist*. Available at: <https://www.economist.com/leaders/2025/10/23/why-china-is-winning-the-trade-war>

It is precisely within this context that the tokenisation of RWA has been a transformative alternative. It is changing how value moves in the global market by using blockchain technology to turn real assets into digital tokens. These tokens are transparent, easy to divide, and can move across borders quickly. Blockchain works as a shared online record that cannot be changed. It replaces many slow and centralised systems, such as banks, and helps make transactions faster and safer. RWA tokenisation turns real assets into sources of liquidity and value. It helps solve the long-standing challenge of where capital originates and how it can be efficiently used.

Hong Kong is rapidly positioning itself as a leading hub for RWA tokenisation in Asia. Through the integration of distributed ledger technology, smart contracts, and regulatory clarity, the region is setting precedents for compliant and scalable asset digitalisation. As case studies in tokenised trade finance, real estate, and new energy industry demonstrate, RWA tokenisation not only improves capital efficiency but also bridges the gap between traditional finance and decentralised finance.

Looking ahead, the tokenisation of RWAs has the potential to fundamentally reshape cross-border financing dynamics for Chinese and Global enterprises. By reducing reliance on collateral-heavy lending, shortening capital circulation cycles, and increasing investor participation, it offers a structural solution to the long-standing liquidity bottlenecks that have constrained overseas expansion. While traditional financing models struggle to keep pace with the global ambitions of Asian enterprises, RWA tokenisation provides a forward-looking mechanism that aligns financial innovation with real economic activity, which unlocks a new era of inclusive, efficient, and borderless capital formation in Asia and beyond.